

ConstructPro ERP

Sprint 7 Frontend Feature Implementation Plan

4-Person Task Breakdown Based on Current Repository Status

1. Current Frontend Status

The current frontend repository already provides the base architecture needed for Sprint 7 integration work. The develop branch contains a Next.js application with shared dashboard routing, authentication foundations, reusable components, API services, Redux state management, and dedicated dashboard modules.

Area	Current Status
Framework	Next.js 16, React 19, TypeScript
State and Forms	Redux Toolkit, React Hook Form, Zod
API Layer	Axios-based service layer already present
Styling	Tailwind CSS with a shared design system
Dashboard Routes	Users, Leads, Quotations, Projects, Finance, Documents, Analytics
Auth	Login, token refresh/session restore foundation already implemented
Existing Services	Finance, Dashboard KPI, Analytics, and AI Forecasting API services already exist

2. Sprint 7 Frontend Goal

Sprint 7 should focus on completing and integrating the existing frontend features rather than rebuilding the dashboard structure. The target is to connect all dashboard modules to real backend APIs, complete role-based workflows, standardize user feedback states, and make the full ERP flow usable from the frontend.

Sprint 7 Theme: Connect, Complete, Integrate, and Polish

Priority	Frontend Focus
Priority 1	Authentication protection, role-based navigation, API contracts, Users, Leads, Quotations, Projects, Finance
Priority 2	Documents, Dashboard KPIs, Analytics charts, Calendar integration, Client Portal
Priority 3	AI prediction UI, PDF actions, shared states, responsive and accessibility polish

3. Person 1 - Authentication, Users, Leads and Calendar

Primary ownership: Admin and Sales workflows.

- Finalize protected dashboard access, login/logout, token refresh, and session restoration behavior.
- Apply role-based navigation so users only see modules allowed for their role.
- Connect User Management to backend APIs: list, create, edit, role assignment, activate/deactivate.
- Connect Lead Management to backend APIs: list, create, edit, assign sales manager, and update status.
- Add search, filters, pagination, loading, empty, error, success, and validation states to lead/user pages.
- Add meeting scheduling UI from lead or project context and connect it to the Google Calendar backend integration.
- Create or complete usersApi.ts, leadsApi.ts, and calendarApi.ts where needed.

Suggested Issue	Scope
FE-01	Complete protected auth and role-based dashboard navigation
FE-02	Connect User Management page to backend APIs
FE-03	Connect Lead Management page to backend APIs
FE-04	Add meeting scheduling UI for Google Calendar integration

4. Person 2 - Quotations and Project Conversion

Primary ownership: Quotation lifecycle from creation through approval and conversion.

- Connect quotation list and detail views to real backend list/detail endpoints.
- Implement lead selection when creating a quotation instead of manual ID entry.
- Complete quotation item add/edit/remove interactions and automatic total preview.
- Show and handle quotation states such as Draft, Sent, Approved, Rejected, Revised, and Converted.
- Implement role-aware approval, rejection, revision, and project conversion actions.
- Handle invalid status transitions, duplicate conversion, missing lead, backend unavailable, and retry states.
- Add quotation PDF view/download actions.
- Create or complete quotationsApi.ts, quotationItemsApi.ts, and projectConversionApi.ts as required.

Suggested Issue	Scope
FE-05	Connect Quotation Management page to backend APIs
FE-06	Implement quotation item form and total calculation UI
FE-07	Implement quotation approval, rejection and revision UI
FE-08	Implement quotation-to-project conversion UI and PDF action

5. Person 3 - Projects, Milestones, Documents and Client Portal

Primary ownership: Project execution and client-facing project information.

- Connect the Projects page to backend list and detail endpoints.
- Display project overview information including client, budget, assigned manager, dates, status, and progress.
- Implement milestone create/edit/status/progress workflows.
- Implement task list, assignment, status, and due date interactions if exposed by the backend.
- Display overall project progress calculated from backend milestone/task data.
- Connect Document Management to upload, categorize, list, view/download, and delete endpoints.
- Connect Cloudinary-backed upload flow through the backend rather than direct client-only handling.
- Implement read-only Client Portal views for a client's own quotations, projects, milestones, progress, and documents.
- Create or complete projectsApi.ts, milestonesApi.ts, tasksApi.ts, and documentsApi.ts as needed.

Suggested Issue	Scope
FE-09	Connect Projects page to backend APIs
FE-10	Implement milestone and task management UI
FE-11	Connect Document Management page with upload/download actions
FE-12	Implement client portal read-only views

6. Person 4 - Finance, Dashboard, Analytics and AI

Primary ownership: Finance and management reporting experience.

- Complete Finance integration for invoice list, invoice details, invoice creation, and payment recording.
- Refresh paid amount, outstanding balance, and invoice status immediately after successful payment.
- Add invoice PDF view/download actions and clear business-rule validation feedback.
- Connect dashboard KPI cards to real revenue, payment, project, lead, and quotation data.
- Complete analytics charts for revenue, payment trends, project progress, and sales conversion metrics.
- Complete AI prediction integration for project risk, payment-delay risk, revenue trend, explanation, and recommended action.
- Standardize loading, empty, unavailable, success, and error states across dashboard modules.
- Finalize API response mapping in financeApi.ts, dashboardApi.ts, analyticsApi.ts, and aiForecastingApi.ts.

Suggested Issue	Scope
FE-13	Complete Finance invoice and payment frontend integration
FE-14	Connect dashboard KPI cards to backend data
FE-15	Complete analytics charts and report views
FE-16	Complete AI risk prediction frontend integration
FE-17	Standardize shared dashboard loading, empty and error states

7. Ownership Summary

Person	Main Pages / Areas	Responsibility
Person 1	Auth, Users, Leads, Calendar	Admin and Sales flows
Person 2	Quotations	Quotation generation, approval and project conversion
Person 3	Projects, Documents, Client Portal	Project operations and client read-only access
Person 4	Finance, Dashboard, Analytics, AI	Finance, reporting and management insights

8. Cross-Person Dependencies

1. Person 1 should finalize authentication and role behavior early because all protected dashboard modules depend on it.
2. Person 2 depends on Lead data from Person 1 for quotation creation and on Project backend behavior for conversion.
3. Person 3 depends on quotation-to-project conversion from Person 2 for end-to-end project flows.
4. Person 4 depends on project and finance data generated by upstream modules for realistic KPI and analytics outputs.
5. All four persons should agree on shared API error handling, loading/empty states, and type conventions before large integration PRs are merged.

9. Recommended Branch Strategy

Person	Example Branches
Person 1	feature/FE-01-auth-role-navigation; feature/FE-02-user-management-ui; feature/FE-03-lead-management-ui; feature/FE-04-calendar-scheduling-ui
Person 2	feature/FE-05-quotation-api-integration; feature/FE-06-quotation-items-ui; feature/FE-07-quotation-approval-ui; feature/FE-08-project-conversion-pdf-ui
Person 3	feature/FE-09-projects-api-integration; feature/FE-10-milestones-tasks-ui; feature/FE-11-document-management-ui; feature/FE-12-client-portal-ui
Person 4	feature/FE-13-finance-api-integration; feature/FE-14-dashboard-kpi-integration; feature/FE-15-analytics-charts-integration; feature/FE-16-ai-risk-prediction-ui; feature/FE-17-shared-dashboard-states

10. Sprint 7 Frontend Completion Criteria

6. Admin can log in and manage users and roles.
7. Sales Manager can create and manage leads and schedule meetings.
8. Sales Manager can create quotations from leads with line items and correct totals.
9. Management can approve/reject quotations and convert approved quotations to projects.
10. Project Manager can view projects, update milestones/tasks, and manage documents.
11. Accountant can create/view invoices and record payments with correct balance/status refresh.
12. Dashboard KPI cards use real backend data.
13. Analytics charts and AI predictions display real backend results.
14. Client Portal exposes only the authenticated client's permitted read-only data.
15. All major pages have consistent loading, empty, error, success, and validation states.
16. Frontend lint, format check, tests, and build pass before merge to develop.